3.1 Measure and integration: problems

Spend at least 30 minutes on each problem (45 minutes on Qualifying-exam level problems) before reading solutions.md. Write full proofs. Throughout, m is Lebesgue measure on $\mathbb{R}$ (or $\mathbb{R}^n$), and "a.e." means "almost everywhere".

Warm-up

1. (a) Show that the intersection of any family of σ-algebras on X is a σ-algebra.

(b) Give an example of two σ-algebras on $X = \{1, 2, 3\}$ whose union is not a σ-algebra.

2. Let μ be a measure and $E_1 \supseteq E_2 \supseteq \cdots$ measurable sets. Show that if $\mu(E_1) < \infty$ then $\mu\left(\bigcap_n E_n\right) = \lim_n \mu(E_n)$, and give an example showing that the hypothesis $\mu(E_1) < \infty$ cannot be dropped.

3. Compute, with justification,

$$\lim_{n\to\infty} \int_0^n \left(1 - \frac{x}{n}\right)^n dx.$$

4. (a) Show that every countable subset of $\mathbb{R}$ has Lebesgue measure zero.

(b) Show that the Cantor middle-thirds set C has measure zero and is uncountable.

5. Let $f \geq 0$ be measurable with $\int f \, d\mu = 0$. Prove that $f = 0$ μ-a.e.

6. (Markov's inequality.) Let f be measurable and $\alpha > 0$. Prove $\mu(\{|f| \geq \alpha\}) \leq \alpha^{-1} \int |f| \, d\mu$. Deduce that every $f \in L^1(\mu)$ is finite a.e.

Core

7. (Vitali.) Construct a subset of $[0, 1]$ that is not Lebesgue measurable, and prove that it is not measurable. State where you used the axiom of choice.

8. Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $m(E) < \infty$ and let $\varepsilon > 0$. Using only the definition of Lebesgue outer measure (as an infimum over countable covers by open intervals) and Carathéodory measurability, show that:

(a) there is an open set $U \supseteq E$ with $m(U \setminus E) < \varepsilon$;
(b) there is a compact set $K \subseteq E$ with $m(E \setminus K) < \varepsilon$.

9. (Egorov.) Let $\mu(X) < \infty$ and let $f_n, f : X \to \mathbb{R}$ be measurable with $f_n \to f$ a.e. Prove that for every $\varepsilon > 0$ there is a measurable E with $\mu(E) < \varepsilon$ such that $f_n \to f$ uniformly on $X \setminus E$. Show by example that the hypothesis $\mu(X) < \infty$ is needed.

10. (a) Give a sequence of non-negative functions f_n on $\mathbb{R}$ for which the inequality in Fatou's lemma is strict.

(b) Derive the dominated convergence theorem from Fatou's lemma.

11. Prove that

$$\lim_{n\to\infty} \int_0^1 \frac{n x^{n-1}}{1 + x} \, dx = \frac{1}{2}.$$

12. (a) Construct a sequence f_n on $[0, 1]$ with $\|f_n\|_1 \to 0$ but $f_n(x)$ convergent for no $x \in [0, 1]$.

(b) Prove that if $f_n \to f$ in measure, then some subsequence converges to f a.e.

13. Let $f(x,y) = \dfrac{x^2-y^2}{(x^2+y^2)^2}$ on $(0,1]^2$. Compute both iterated integrals $\int_0^1\int_0^1 f\,dy\,dx$ and $\int_0^1\int_0^1 f\,dx\,dy$, and explain why this does not contradict Fubini's theorem.

14. Let $f \in L^1(\mathbb{R})$. Prove that $\lim_{h\to 0}\int_{\mathbb{R}}|f(x+h)-f(x)|\,dx = 0$. You may use that $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$. Show that the analogous statement fails in $L^\infty(\mathbb{R})$.

15. Let $\mu(X) < \infty$.

(a) Show that if $1 \le p < q \le \infty$ then $L^q(\mu) \subseteq L^p(\mu)$ and $\|f\|_p \le \mu(X)^{1/p-1/q}\|f\|_q$.
(b) Show that $\lim_{p\to\infty}\|f\|_p = \|f\|_\infty$ for every $f \in L^\infty(\mu)$.

16. (Riesz–Fischer.) Prove that $L^p(\mu)$ is complete for $1 \le p < \infty$.

17. Let $f \in L^1(\mathbb{R})$ be real-valued and define the signed measure $\nu(E) = \int_E f\,dm$. Find a Hahn decomposition for ν, its Jordan decomposition, and prove that $|\nu|(E) = \int_E |f|\,dm$.

18. Let m be Lebesgue measure on the Borel sets of $[0,1]$ and c counting measure on the same σ-algebra. Show that $m \ll c$ but that there is no measurable f with $m(E) = \int_E f\,dc$ for all Borel E. Which hypothesis of the Radon–Nikodym theorem fails?

19. (Borel–Cantelli.) (a) Prove that if $\sum_n \mu(E_n) < \infty$ then almost every point lies in only finitely many E_n.

(b) Deduce that for almost every $x \in \mathbb{R}$ there are only finitely many pairs of integers (p,q) with $q \ge 1$ and $\left|x - \frac{p}{q}\right| < \frac{1}{q^3}$.

Qualifying-exam level

20. Let $f \in L^1(\mathbb{R})$. Prove that $\sum_{n\in\mathbb{Z}}|f(x+n)| < \infty$ for almost every x, and that the function $F(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ satisfies $\int_0^1 F(x)\,dx = \int_{\mathbb{R}} f(x)\,dx$.

21. Let $f \in L^1[0,2\pi]$. Prove that

$$\lim_{n\to\infty}\int_0^{2\pi} f(x)\,|\sin nx|\,dx = \frac{2}{\pi}\int_0^{2\pi} f(x)\,dx.$$

22. Let $1 \le p < \infty$, and let $f_n, f \in L^p(\mu)$ with $f_n \to f$ a.e. and $\|f_n\|_p \to \|f\|_p$. Prove that $\|f_n - f\|_p \to 0$.

23. Let $f_n \in L^2[0,1]$ with $\|f_n\|_2 \le 1$ for all n, and suppose $f_n \to f$ a.e. Prove that $f \in L^2[0,1]$ with $\|f\|_2 \le 1$, and that $\int_0^1 f_n g \to \int_0^1 fg$ for every $g \in L^2[0,1]$.

24. Let $E \subseteq \mathbb{R}$ be Lebesgue measurable and suppose there is a constant $c < 1$ such that $m(E \cap I) \le c\,m(I)$ for every bounded open interval I. Prove that $m(E) = 0$.

25. Let $F : [0,1] \to [0,1]$ be the Cantor function and μ_F the associated Lebesgue–Stieltjes measure on $[0,1]$ (so $\mu_F([0,x]) = F(x)$).

(a) Prove that $\mu_F \perp m$.
(b) Prove that F is not absolutely continuous.
(c) Compute $\int_0^1 F(x)\,dx$.